

Yixiu Xiao

PhD Candidate in Mathematics, SJTU
Visiting Research Student, UNSW Sydney



Personal details

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Education

Apr 2025–present **Visiting Research Student (Joint Training Program), The University of New South Wales (UNSW Sydney), Sydney, Australia**
Supervisor: Prof. Igor E. Shparlinski.

Sep 2021–expected Mar 2027 **Ph.D. Candidate, Mathematics, Shanghai Jiao Tong University, Shanghai, China**
Supervisor: Prof. Hongze Li.

Sep 2017–Jun 2021 **B.Sc. in Mathematics and Applied Mathematics, Northwest University, Xi'an, China**

Research Interests

Analytic Number Theory

- Exponential sums, character sums, and bilinear forms
- Kloosterman and Salié sums
- Sieve methods, large sieve inequalities, and the distribution of primes
- Equidistribution modulo one and discrepancy theory
- Diophantine equations and the Hardy–Littlewood circle method

Current Research Directions

- Sign changes of Kloosterman sums over square-free almost-prime moduli
- Moment estimates and discrepancy for sums of square roots modulo one
- Large sieve inequalities for quadratic character sums over short intervals
- Shifted bilinear sums of Salié sums and modular square roots of shifted primes
- Least non-trivial zeros of cubic forms and systems of additive cubic equations

Publications

Published

Least Zero of a Cubic Form, with Hongze Li. *Frontiers of Mathematics*, 2025. DOI: 10.1007/s11464-024-0113-6.

Least Zero of Pairs of Additive Cubic Equations, with Hongze Li. *Forum Mathematicum*, 2026. DOI: 10.1515/forum-2025-0367.

Preprints (arXiv)

Kloosterman sign changes with moduli having at most five prime factors. *arXiv*: 2607.10862.

Moment Estimates and Discrepancy for Sums of Square Roots Modulo One. *arXiv*: 2606.28986.

Large sieve inequality for sums of Legendre symbols over short intervals, with Marc Munsch, Igor E. Shparlinski, and Yu-Chen Sun. *arXiv*: 2604.23661.

Shifted bilinear sums of Salié sums and the distribution of modular square roots of shifted primes, with Igor E. Shparlinski. *arXiv*: 2601.10113.

Scholarships

2024 China Scholarship Council (CSC), State Scholarship Fund (Visiting PhD Student to Australia), 24 months. Covers international airfare, living stipend, and health insurance.

Talks

Diophantine equations and cubic forms, Prime Pals / Number Theory Group Seminar (internal), UNSW Canberra, Canberra, 9 Sep 2025.

Least zero of pairs of additive cubic equations, AustMS Annual Meeting, La Trobe University, Melbourne, 10 Dec 2025.

Research Experience

Research Assistant, Prof. Hongze Li, Shanghai Jiao Tong University, 2021–present.

Conducted research in analytic number theory, including exponential sums, character sums, Diophantine equations, and related problems in number theory. Participated in literature study, mathematical discussions, proof development, and manuscript preparation.

Research Assistant, School of Mathematics and Statistics, UNSW Sydney, 7 April 2026–present. Supervisor: Prof. Igor E. Shparlinski.

Teaching

Mathematical Analysis, Teaching Assistant, Shanghai Jiao Tong University, Fall terms (2021–2023).

Elementary Number Theory (Summer Term), Teaching Assistant, Shanghai Jiao Tong University, 2022–2024.

MATH8000 Student Support (Student Supervision), UNSW Sydney, Term 3 2025.

MATH8000-Student Support (CTP), UNSW Sydney, Term 1 2026–present.

MATH1141 Higher Mathematics 1A, Routine Marker, UNSW Sydney, Term 1 2026–present.